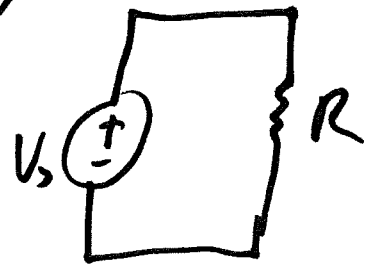


9.28)

$$R = 20$$

$$V_s = 120 \cos(377t + 37^\circ)$$

$$I = ?$$



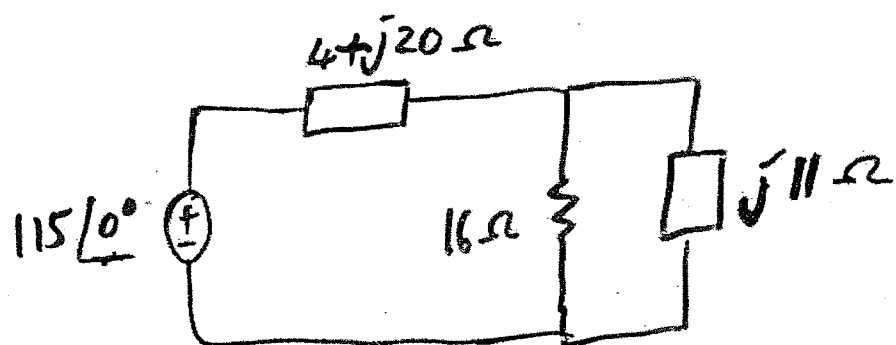
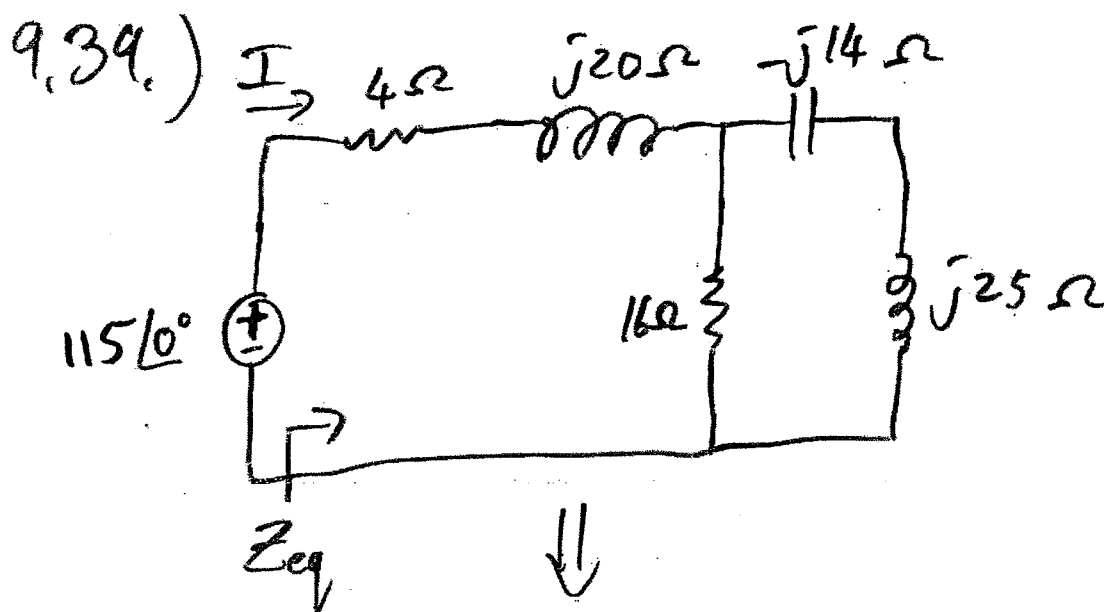
$$R = \frac{V}{I}$$

$$20 = \frac{120 \angle -37}{I}$$

$$\therefore \underline{I} = \frac{120 \angle 37}{20 \angle 0^\circ}$$

$$= 6 \angle 37^\circ$$

$$\underline{I(t)} = 6 \cos(377t + 37^\circ)$$



$$\begin{aligned}
 \therefore Z_{eq} &= (16 \parallel j11) + 4 + j20 \\
 &= \frac{(16 \angle 0^\circ)(11 \angle 90^\circ)}{16 + j11} + 4 + j20 \\
 &= \frac{176 \angle 90^\circ}{19.42 \angle 34.51^\circ} + 4 + j20 \\
 &= 9.06 \angle 55.49^\circ + 4 + j20 \\
 &= 5.13 + j7.47 + 4 + j20 \\
 &= 9.13 + (27.47j) \\
 &= \underline{28.95 \angle 71.62^\circ}
 \end{aligned}$$

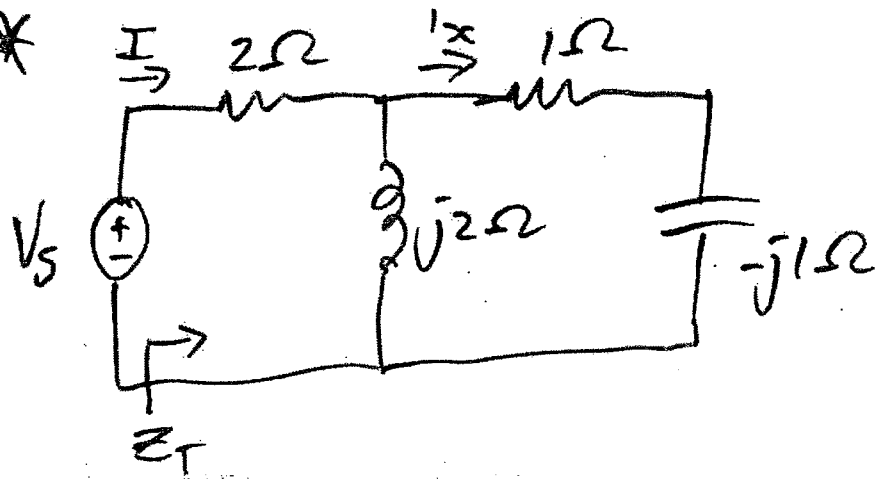
$$I = \frac{V}{R}$$

$$= \frac{115 \angle 0}{28,95 \angle 71,62}$$

$$= 3,97 \angle -71,62$$

$$= 3,97 \cos(10t - 71,62)$$

1.49.)*



Determine V_s

$$i_x = 8 \sin 200t = 8 \cos (200t - 90) \text{ A}$$

1) Calculate Z_T :

$$Z_T = (1 - j) \parallel j2 + 2$$

$$= 2 + \frac{(1 - j)(j2)}{1 - j + j2}$$

$$= 2 + \frac{(\sqrt{2} \angle -45^\circ)(\sqrt{2} \angle 90^\circ)}{1 - j}$$

$$= 2 + \frac{2\sqrt{2} \angle -45^\circ}{\sqrt{2} \angle -45^\circ}$$

$$= 4 \angle 0^\circ \Omega$$

3) Using current division:

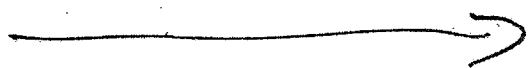
$$I_x = I \left(\frac{j2}{j2 + 1 - j} \right)$$

$$8 \angle -90^\circ = I \left(\frac{j2}{1 + j} \right)$$

$$8 \angle -90^\circ = I \left(\frac{2 \angle 90^\circ}{\sqrt{2} \angle 45^\circ} \right)$$

$$8 \angle -90^\circ = I (1,41 \angle 45^\circ)$$

$$I = 5,67 \angle -135^\circ \text{ A}$$



4) Calculate V_s :

$$Z_T = \frac{V_s}{I}$$

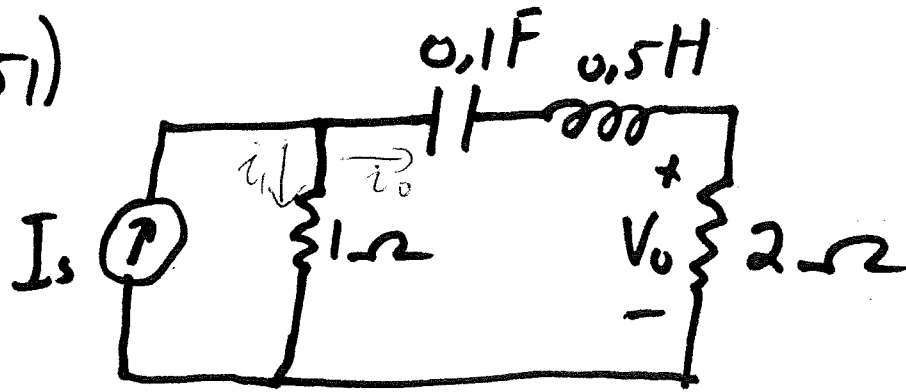
$$V_s = I Z_T$$

$$V_s = (5,67 \angle -135^\circ) (4 \angle 0^\circ)$$

$$V_s = 22,7 \angle -135^\circ$$

$$V_s = 22,7 \cos(200t - 135^\circ)$$

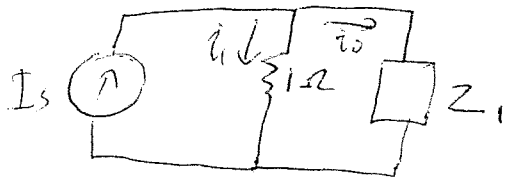
9.51)



$$V_o = 90 \cos 2t \text{ V}$$

$$I_s = ?$$

$$i_o = \frac{V_o}{2} = \frac{90 \angle 0^\circ}{2} = \underline{45 \angle 0^\circ}$$



$$Z_1 = 2 + \frac{1}{j\omega C} + j\omega L$$

$$= 2 - j5 + j$$

$$\underline{Z_1 = 2 - j4}$$

$$V_{Z_1} = i_o Z_1 = (45)(2 - j4)$$

$$\underline{V_{Z_1} = 90 - j180 = V_{1-2}}$$

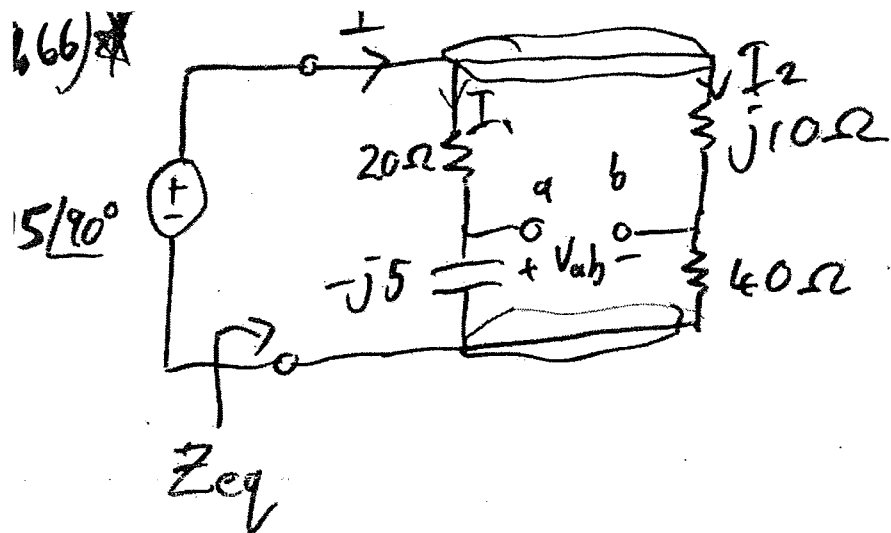
$$i_1 = \frac{V_{Z_1}}{1} = 90 - j180$$

$$I_s = i_o + i_1$$

$$= 135 - j180$$

$$= 225 \angle -53.13^\circ$$

$$\underline{= 225 \cos(2t - 53.13^\circ) \text{ A}}$$



$$Z_{eq} = ?$$

$$V_{ab} = ?$$

1) Z_{eq} :

$$Z_{eq} = (20 - j5) \parallel (40 + j10)$$

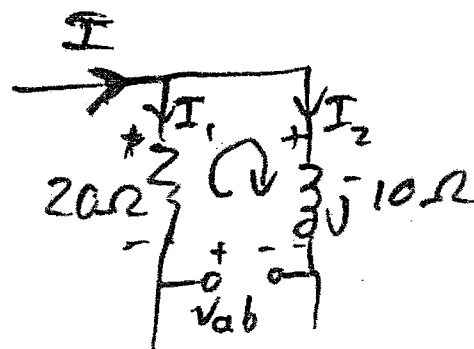
$$= \frac{(20 - j5)(40 + j10)}{60 + j5}$$

$$= 14,069 - j1,17 \Omega = 14,12 \angle -4,76^\circ$$

2) V_{ab} :

$$* I = \frac{V}{Z_{eq}} = \frac{115 \angle 90^\circ}{14,12 \angle -4,76^\circ}$$

$$= 8,14 \angle 94,76^\circ$$



③ KVL:

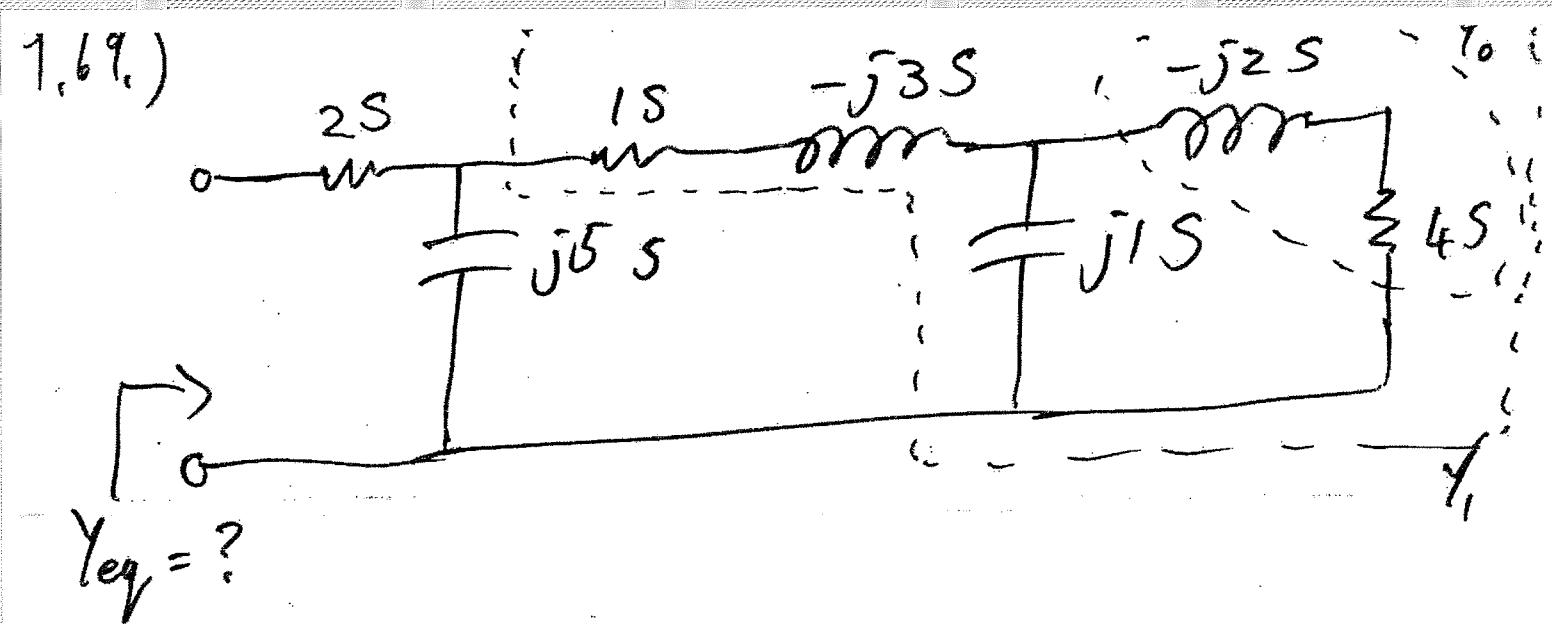
$$V_{ab} = -20I_1 + (j10)I_2$$

$$V_{ab} = \frac{-150}{12 + j} I$$

$$V_{ab} = 101,38 \angle 89,97^\circ V$$

$$* I_1 = I \frac{40 + j10}{60 + j5}$$

$$* I_2 = I \frac{20 - j5}{60 + j5}$$



$$① \quad \frac{1}{Y_0} = \frac{1}{4} + \frac{1}{-j2} = \frac{1}{4} (1 + j2)$$

$$Y_0 = \frac{4}{1 + j2} = 0,8 - j1,6$$

$$② \quad \frac{1}{Y_1} = \frac{1}{1} + \frac{1}{-j3} + \frac{1}{0,8 - j0,6} = (1) + j(0,33) + \frac{1}{0,8 - j0,6}$$

$$Y_1 = 0,447 + (-j0,2271)$$

$$③ \quad \frac{1}{Y_{eq}} = \frac{1}{2} + \frac{1}{j5 + 0,44 - j0,227} = \frac{1}{2} + \frac{1}{0,44 - j4,77}$$

$$Y_{eq} = (1,66 + j0,66)S$$